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🎓 Agentic AI Fundamentals Quiz

🧠 Section A: Multiple Choice Questions (MCQs)

1. What is Agentic AI?
A. AI that only performs predefined tasks
B. AI systems that act autonomously to achieve goals
C. AI used only for data storage
D. AI that requires constant human supervision

2. Which of the following best describes an "agent" in AI?
A. A database
B. A program that perceives and acts upon an environment
C. A neural network only
D. A hardware device

3. What is the primary goal of an AI agent?
A. To maximize randomness
B. To achieve a defined objective or goal
C. To reduce memory usage
D. To store large datasets

4. Which component is responsible for deciding actions in an agent?
A. Sensor
B. Actuator
C. Policy/Decision-making module
D. Memory storage

5. In Agentic AI, "planning" refers to:
A. Collecting data
B. Creating a sequence of actions to achieve a goal

C. Training a model
D. Visualizing results

6. Which of the following is NOT a typical agent role?
A. Perception
B. Reasoning
C. Execution
D. Compilation

7. Reasoning in AI agents helps to:
A. Store information
B. Make logical decisions based on data
C. Increase hardware speed
D. Reduce dataset size

8. What does “execution” mean in Agentic AI?
A. Designing algorithms
B. Running planned actions in the environment
C. Training a model
D. Debugging code

9. Which type of agent uses condition-action rules?
A. Learning agent
B. Model-based agent
C. Simple reflex agent
D. Utility-based agent

10. Multi-agent systems involve:
A. Only one AI system
B. Multiple agents interacting with each other
C. Only human interaction
D. Only hardware devices

✍️ Section B: Short Answer Questions

11. Define Agentic AI in your own words.

Answer:-

Agentic AI refers to AI systems that can perceive their environment, reason about information, plan actions, and act autonomously to achieve specific goals.

12. List any three roles of an AI agent.

Answer:-

- Perception – collecting information from the environment
- Reasoning – analyzing information to make decisions
- Execution – performing actions in the environment

13. What is the difference between planning and execution?

Answer:-

Planning	Execution
Deciding **what actions should be taken**	**Performing** the chosen actions
Happens before acting	Happens during action
Example: choosing route	Example: driving on that route

14. Explain the importance of reasoning in AI agents.

Answer:-

Reasoning allows AI agents to analyze data, evaluate options, and make logical decisions instead of blindly following rules. It helps the agent adapt to new situations.
Example:
A chatbot reasoning about user intent before replying.

15. What are sensors and actuators in an agent system?

Answer:-

Sensors
Devices or mechanisms that collect information from the environment.
Examples:

- Cameras

- Microphones
- Temperature sensors

Actuators

Components that perform actions in the environment.

Examples:

- Motors
- Speakers
- Robot arms

🔍 Section C: Application-Based Questions

16.

A delivery robot must pick up packages and deliver them efficiently.

- Identify the **agent roles** involved.

Answer:-

Roles involved:

1. Perception – detecting obstacles and packages
2. Planning – choosing the best delivery route
3. Reasoning – deciding alternative paths if blocked
4. Execution – moving and delivering packages

17.

An AI chatbot analyzes user queries and responds intelligently.

- Explain how **reasoning** is used.

Answer:-

A chatbot:

1. Analyzes user input
2. Understands intent
3. Chooses the best response

Example:

User:

"I forgot my password."

Reasoning:

- Detect intent → password reset

- Provide reset instructions.

18.
Design a simple agent for a **smart home system**.

- What would be its inputs, outputs, and goals?

Answer:-

Goal:
Automate home environment for comfort and efficiency.
Inputs (Sensors):
- Temperature sensors
- Motion detectors
- Light sensors
- User voice commands

Outputs (Actions):
- Turn lights on/off
- Adjust AC/heater
- Lock doors
- Send notifications

19.
Explain how **planning + reasoning + execution** work together in a self-driving car.

Answer:-

Planning
- Determine route to destination.

Reasoning
- Analyze traffic lights, pedestrians, road signs.

Execution
- Accelerate, brake, steer.

Example flow:
Environment → Sensors → Reasoning → Planning → Execution → Movement.

20.
Why is autonomy important in Agentic AI? Give a real-world example.

Answer:-

Autonomy allows AI systems to operate without constant human control, making them faster and more efficient.

Example:

- Autonomous delivery robots
- Self-driving cars
- AI customer support agents

These systems make decisions and act independently.

⚙ Section D: Higher-Order Thinking

21. Compare Agentic AI with traditional rule-based systems.

Answer:-

Agentic AI	Rule-Based Systems
Adaptive and autonomous	Fixed rules
Uses reasoning and planning	Uses if-else conditions
Can learn and improve	Cannot adapt easily

Example:

- Agentic AI → self-driving car
- Rule-based → basic thermostat.

22. What challenges might arise in multi-agent systems?

Answer:-

1. Coordination problems
2. Communication between agents
3. Conflicting goals
4. Scalability issues
5. Security risks

Example: traffic coordination among autonomous cars.

23. How can Agentic AI improve real-world decision-making?

Answer:-

Agentic AI can:

- Analyze large amounts of data
- Predict outcomes
- Adapt to changing environments
- Make fast decisions

Example:

- Healthcare diagnosis systems
- Financial trading agents
- Smart traffic management.

24. Explain how feedback helps improve agent performance.

Answer:-

Feedback helps agents learn from previous actions.
Process:
Action → Result → Feedback → Improvement.
Example:

- Reinforcement learning in robotics.

25. Can Agentic AI function without human input? Justify your answer.

Answer:-

Yes, but within defined boundaries.
Humans usually:

- Set goals
- Define constraints
- Monitor performance

Once deployed, the agent can operate autonomously.
Example:

- Autonomous drones performing surveillance.